

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Chandubhai S Patel Institute of Technology

1st Semester B.Tech.

University Exam Nov-Dec 2011

MA-101 Engineering Mathematics-I

Date: 05/12/'11

Time: 01:30pm to 04:30pm

Maximum Marks: 70

Instruction:

1. This question paper contains two sections **Section-I** and **Section-II**.
2. **Section-I** carries 35 and **Section-II** carries 35 marks.
3. Section wise separate answer book to be used.
4. Figure to the right indicates the distribution of full marks for the respective question.

Section-I

Q.1 Attempt any Three

[12]

- (i) Check consistency of the following systems and solve by Gauss Elimination method:

$$x + 2y + z = 4, 2x + y = 3, x + z = 2$$

- (ii) Find the roots of the equation $z^4 + 1 = 0$

(iii) Find inverse of
$$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

- (iv) Find radius and interval of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{\sqrt{2n^2 - n}}$

- (v) If $i^{i^{\dots\infty}} = A + iB$, prove that

(a) $\tan \frac{\pi A}{2} = \frac{B}{A}$ and (b) $A^2 + B^2 = e^{-\pi B}$

Q.2 Attempt any Four

[12]

- (i) Prove that $\frac{(\cos 2\theta - i \sin 2\theta)^5 (\cos 3\theta + i \sin 3\theta)^6}{(\cos 4\theta + i \sin 4\theta)^7 (\cos \theta - i \sin \theta)^8} = (\cos 12\theta - i \sin 12\theta)$

- (ii) Discuss the convergence of $\sum_{n=0}^{\infty} \left(\frac{n-2}{n} \right)^n$

(iii) Find rank and nullity of
$$\begin{bmatrix} 2 & 3 & 4 & 5 \\ 4 & 5 & 5 & 6 \\ 7 & 8 & 9 & 10 \end{bmatrix}$$

- (iv) Evaluate $\frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \dots$

- (v) Separate real and imaginary part of $\tan(x - iy)$.

Q.3 (a) Determine the following series convergence or divergence

[6]

(i) $\sum_{n=0}^{\infty} \frac{n!}{3^n}$ (ii) $\sum_{n=0}^{\infty} \frac{2n-1}{n(n+1)(n+2)}$

OR

Q.3 (a)(i) If the rank of the matrix $\begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -1 & 0 & \lambda \end{bmatrix}$ is 2, then find the value of λ . [3]

(ii) Prove that $(1 + i\sqrt{3})^n + (1 - i\sqrt{3})^n = 2^{n+1} \cos \frac{n\pi}{3}$ [3]

Q.3 (b) For what value of k , the equations $x + 2y + z = 3$, $x + y + z = k$ and $3x + y + 3z = k^2$ are consistent? Solve the equations for all possible values of k . [5]

Section-II

Q.1 Attempt any Three

[12]

(i) If $u = x \sin \left(\frac{x}{y} \right) + \log \left(\frac{y}{x} + \frac{x}{y} \right)$, then find $xu_x + yu_y$ and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.

(ii) If $u = \log_e(x^3 + y^3 + z^3 - 3xyz)$, prove that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = \frac{-9}{(x+y+z)^2}$.

(iii) Expand $\log_e \cos x$ in power of $\left(x - \frac{\pi}{3} \right)$ and hence evaluate $\log_e \cos 59^\circ$.

(iv) If $y\sqrt{1+x^2} = \log_e(x + \sqrt{1+x^2})$,

prove that $(1+x^2)y_{n+1} + (2n+1)xy_n + n^2y_{n-1} = 0$.

(v) A rectangular box open at the top is to have volume of 32 cubic feet. Find the dimension of the box requiring minimum material for its construction.

Q.2 Attempt any Four

[12]

(i) If $x = r \cos \theta$ and $y = r \sin \theta$, find Jacobian $\frac{\partial(r, \theta)}{\partial(x, y)}$.

(ii) If $u = f(x+ct) + g(x-ct)$, show that $u_{tt} = c^2u_{xx}$.

(iii) If the kinetic energy $T = \frac{1}{2}mv^2$, find approximately the change in T as m changes from 49 to 1.02 percent and v changes from 1600 to 1590.

(iv) Find $\frac{dy}{dx}$ for an implicit function, $x^3 + y^3 - 3axy = 0$ at $(1, 1)$.

(v) Find Maclaurin's series expansion of $\tan^{-1} x$ up to x^5 .

(vi) Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$.

Q.3 (a) (i) prove that $3 + \frac{1}{28} < \sqrt[3]{28} < 3 + \frac{1}{27}$. [3]

(ii) Find the equation of the tangent plane and normal line to the surface

$x^2 + y^2 + z^2 = 1$ at $\left(\frac{2}{3}, \frac{2}{3}, \frac{1}{3} \right)$. [3]

OR

Q.3 (a) (i) Find n^{th} derivative of $y = \frac{x}{(2x-1)(x+3)}$. [3]

(ii) If $f(x) = e^x$ and $g(x) = e^{-x}$ defined on $[a, b]$, using Cauchy's Mean value theorem,

show that there exist some $c \in (a, b)$ such that c is the average of a and b . [3]

Q.3 (b) Find extreme(maximum/minimum) values of $z = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$. [5]